

Hasan Khan

Senior Software Engineer

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Summary

Senior Software Engineer with 8+ years of experience building scalable back-end systems, native APIs in the cloud, and high performance data pipelines for Fortune 100 companies. Expert in Python, AWS, and SQL-based platforms, with experience delivering robust, production-grade systems. Proven ability to lead backend efforts on enterprise-scale applications, resolve cross-team production issues, and deliver impactful features in fast-paced Agile environments. M.S. in Computer Science from Georgia Tech.

Skills

Languages & Frameworks:	Python, TypeScript, Node.js, NestJS, FastAPI, SQLAlchemy, SQL, Pandas, NumPy
Cloud & Infrastructure:	AWS (Lambda, API Gateway, RDS, EC2, DynamoDB, S3, SES, SNS, CloudWatch), Docker, Kubernetes, Git
Databases & Data Engineering:	PostgreSQL, DynamoDB, SQL Server, Redis, ETL Pipelines, NoSQL, Time-Series Data, Data Lakes
Analytics:	Predictive Analytics, Predictive Maintenance, Data Modeling
Architecture & DevOps:	Microservices, Middleware, RESTful APIs, Agile/Scrum

Experience

AT&T

Senior Software Engineer

Aug 2024 – Present

Atlanta, GA (Hybrid)

- Lead backend developer for the Customer Self-Install (CSI) feature in the **myAT&T mobile app** — a production system used by **hundreds of millions of customers** — built using **TypeScript**, **NestJS**, and **Node.js** microservices to support in-home internet activation without technician visits.
- Remediated long-standing **SonarQube** violations and established backend code standards for one of the project's major microservices, enabling a **clean code milestone** and improving maintainability across multiple teams.
- Delivered backend features across a large-scale **NestJS/Node.js** platform while navigating sparse documentation and cross-team blockers within a globally distributed team.
- Implemented backend logic to expose device-level network pause timers and usage metrics across all connected devices within a user's account.

Delta Air Lines

Software Engineer

Oct 2021 – Jan 2024

Atlanta, GA (Hybrid)

- Developed a **Python**-based monitoring and alerting system for Delta's flight training optimizer using **AWS Step Functions**, **SNS**, and **SES**, reducing failure response time from days to hours.
- Built a production-grade support ticket API using **FastAPI**, **AWS Lambda**, and **SES**, enabling seamless in-app issue reporting and improving stakeholder confidence.
- Authored a comprehensive 69-page documentation suite for backend APIs, improving onboarding and long-term maintainability.
- Created backend APIs to exchange metadata between Delta's scheduling optimizer and cloud data stores including **PostgreSQL (RDS)**, **DynamoDB**, and **S3**.

Nike (Contract)

Senior Software Engineer

Dec 2022 – Jun 2023

Boston, MA (Remote)

- Integrated predictive models into production pipelines to support supply chain forecasting and optimization in collaboration with data scientists.
- Built and deployed backend infrastructure using **AWS**, **Docker**, and **Kubernetes** to support scalable forecasting systems across Nike's global supply chain.
- Developed and maintained **Python**-based data pipelines powering demand optimization and logistics analytics workflows.

Material Handling Systems

Software Engineer

May 2019 – Sep 2021

Atlanta, GA

- Enhanced a **Python**-based anomaly detection algorithm to monitor complex time-series machine data, enabling proactive fault detection and reducing operational downtime.
- Identified a **critical flaw** in the central PostgreSQL schema storing raw machine telemetry, then **redesigned it into a star schema layout** to enable downstream analytics, restore operational visibility, and support data warehouse integration.

- Developed a weighted scoring engine to calculate asset health ratings from historical alarms, supporting predictive maintenance for industrial robotics and conveyors.
- Maintained modular data pipeline components that used **SQLAlchemy** to extract, transform, and consolidate telemetry from diverse machine sources.

SimplePart

Jun 2018 – May 2019

Software Developer

Atlanta, GA

- Led the migration of 7 business-critical e-commerce applications to a redesigned release version on the existing **C#**, **.NET**, and **SQL Server** stack, ensuring continuity of systems generating \$100K+ in monthly revenue.
- Refactored legacy **Python** code for a computer vision tool used to extract automotive part images from exploded-view assembly diagrams.
- Modernized front-end components for **WCAG** accessibility compliance using **HTML**, **CSS**, **Bootstrap**, **JavaScript**, and **jQuery**.

CirrusLabs

May 2018 – Jun 2018

Software Engineer Intern

Atlanta, GA

- Built a **GCP**-based time-series ingestion pipeline and implemented a **Redis** caching layer in **Python** to support high-volume welding telemetry for a predictive maintenance system.

Education

Georgia Institute of Technology

Graduated: May 2025

M.S. in Computer Science

Thomas Edison State University

Graduated: Dec 2020

B.A. in Computer Science

Academic Projects

Diabetes Management App

Jan 2024 – Apr 2024

Developed a full-stack health tracking platform using React, FastAPI, and PostgreSQL; implemented secure user authentication, RESTful APIs, and HL7 FHIR-compliant data models for managing glucose, meals, exercise, and medications.

ASL Wordle

Sep 2023 – Dec 2023

Built a browser-based Wordle clone playable using real-time ASL gestures via webcam input; used TensorFlow.js to classify real-time ASL gestures from webcam input and React for frontend interaction.

Stock Portfolio Optimizer

Mar 2022 – May 2022

Created a Python tool to compute optimal asset allocations using historical return data; maximized Sharpe ratio through matrix-based optimization with Pandas and NumPy.

CART Returns Forecasting

Jan 2022 – Mar 2022

Implemented and tuned Classification and Regression Tree (CART) models to predict emerging market index returns; applied cross-validation, feature selection, and statistical evaluation in Python.

ASL Sign Recognizer

Oct 2021 – Dec 2021

Developed a word-level ASL recognition system using Hidden Markov Models; trained on motion sensor time-series data with Viterbi and forward-backward decoding for gesture sequence classification.